



Built to withstand harsh environments, an outdoor-rated enclosure and standard 1000-foot motor cable runs mean the drive provides maximum mounting flexibility

The Perfect solution for demanding applications, such as:

- Lift stations
- Pump stations
- Irrigation
- Other outdoor applications

Neptune-Benson's Variable Frequency Drive (VFD) is suitable for outdoor or indoor installations that require protection against windblown dust and rain or splashing water, IP66-rated variable frequency drives can be installed directly at the equipment location without a protective enclosure. All cast aluminum parts are powder coated with a durable epoxy that can stand up to most corrosive chemicals.

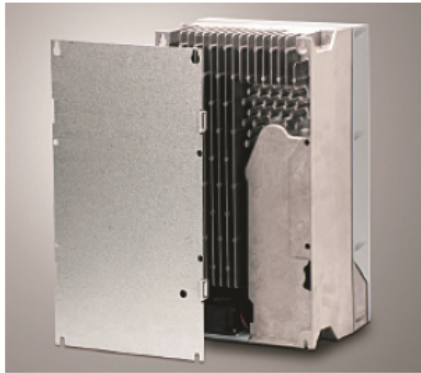
Features	Benefits
• All cast aluminum parts are powder coated with a durable epoxy coating	• No need for separate cover or enclosure*
• Conformal coated circuit boards	• Additional protection in corrosive environments
• All stainless steel screws	• Improved corrosion resistance
• Fan designed to withstand corrosion	• Reliable operation
• Installed near the motor or blower	• Facilitates modular plant design
• 1000' motor cable runs (unshielded)	• Short motor cables reduce EMI/RFI

Reliable	Maximum uptime
• Robust, single enclosure	• Reduced cost and maintenance
• Unique cooling concept with no ambient air flow through electronics housing	• Additional protection in corrosive environments
• Maximum ambient temp. 50° C without derating	• Reliable operation in harsh environments
• Installed near the motor or blower	• No external cooling or over sizing necessary
• 1000' motor cable runs (unshielded)	

User friendly	Simplified operation and lower costs
• Award-winning LCP keypad design	• Reduced space requirements and commissioning time
• Easy installation	• Lower start up costs
• Watertight USB plug can be mounted in the bottom of the drive	• The drive enclosure can remain closed while making setup or programming changes

* For outdoor installations: The drive must be installed under a suitable cover to protect from direct exposure to sun, snow and ice.

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Stainless steel back plate

For panel or wall mounting, a stainless steel back plate is available to direct the air from the fan through the rear heatsink.



Watertight USB plug

A USB plug can be mounted in the bottom of the enclosure, allowing the drive to stay closed while making setup or programming changes using MCT 10 Setup software.

PC software tools

- **MCT 10** – ideal for starting up and servicing the drive
- **MCT 31** – harmonics calculations tool

Mains supply (L1, L2, L3)	
Supply voltage	200-240V +- 10%, 380-480V +-10%, 525-690V +-10%
Supply frequency	50/60 Hz
Displacement Power Factor	(>0.98)
Switching on input supply L1, L2, L3	1-2 times/min.
Output data (U, V, W)	
Output voltage	0-100% of supply
Switching on output	Unlimited
Ramp times	1-3600 sec.
Closed Loop	0-132 Hz
Digital Inputs/outputs	
Programmable digital inputs (std)	6 (two can be used as digital outputs)
General purpose I/O card (option)	Additional 3 digital inputs, 2 digital outputs
Logic	PNP or NPN
Voltage level	0-24 VDC
Digital Inputs/outputs	
Analog inputs (standard)	2
General purpose I/O card (option)	2 additional analog inputs
Advanced analog I/O card (option)	3 additional analog inputs
Modes	Voltage or current
Voltage level	-10 to +10V (scaleable)
Current level	0/4 to 20mA (scaleable)
Pulse Inputs	
Programmable pulse inputs (std)	2 (two digital inputs can be used as pulse inputs)
Voltage level	0-24VDC (PNP positive logic)
Pulse input accuracy	(0.1 - 110kHz)
Analog Inputs	
Programmable analog outputs (std)	1
General purpose I/O card (option)	1 additional analog current output
Advanced analog I/O (option)*	3 additional analog outputs
Current range at analog output	0/4 - 20 mA
Analog outputs	
Programmable relay outputs (std)	2 (240VAC, 2A and 400VAC, 2A)
Relay card (option)	3 additional dry contact relays (240VAC, Form C)
Voltage level	0-24VDC (PNP positive logic)
Pulse input accuracy	(0.1 – 110kHz)
External DC supply	
External 24V DC supply card (option)	Provides backup power for control and option cards
Fieldbus communication	
FC Protocol and Modbus RTU build in (LonWorks, DeviceNet, Profibus modules optional)	
Ambient Temperature	
50°C	

*Advanced analog I/O option card also provides 24V DC backup power for the drive's real-time clock

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